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Objective

The objective of this report is to document the biological evaluation plan (BEP) developed by MED Institute for the Theromics HeatSYNC® Gel. This BEP relies on the device description, intended use, and materials of construction provided by the client, as well as knowledge of current standards to produce a recommended plan for testing and rationale expected to demonstrate biological safety in light of current regulatory expectations. Please note that the assessment was performed by MED Institute, and was not verified by Jordi Labs.

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Sincerely,

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Consulting Report: Biological Evaluation Plan

Device: HeatSYNC® Gel

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Device Description

HeatSYNC® Gel is a yellow, translucent gel of high viscosity that functions as a thermal accelerant during image-guided thermal ablation (i.e., with microwave or radiofrequency ablation systems). It is indicated for use with thermal ablation devices to improve energy conduction and generate greater ablation zone temperatures; it is intended to extend the ablation zone and provide more rapid ablation.

HeatSYNC® Gel may be applied percutaneously, bronchoscopically, intraoperatively, or laparoscopically to target regions of soft tissue. The use of HeatSYNC® Gel does not alter the thermal ablation procedure. If HeatSYNC® Gel is used, it is injected into the desired location 1.5 to 2.0 cm radially from the ablation antenna. With the exception of injecting the HeatSYNC® Gel into the desired location, the ablation procedure is the same. Ablation is performed using a commercially available ablation device per standard of care and according to the instructions for use of the thermal ablation system. HeatSYNC® Gel is inherently visible under ultrasound, but to facilitate visualization of the injection process, HeatSYNC® Gel may be mixed with a commercially available contrast agent.

HeatSYNC® Gel will be supplied sterile in a glass vial; see Figure 1.

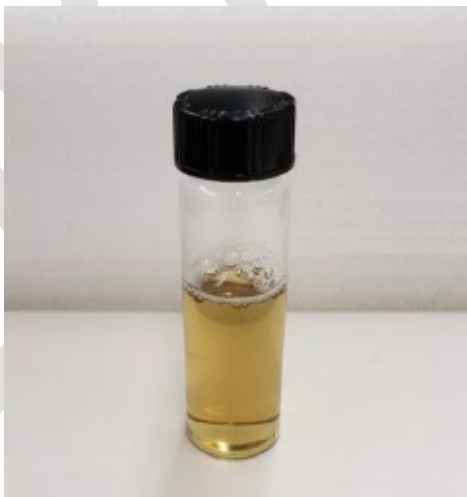


Figure 1. HeatSYNC® Gel (10 mL)

HeatSYNC® Gel is a protein-based gel, primarily composed of human serum albumin. The proprietary formulation is prepared by dissolving citric acid trisodium salt dihydrate and citric acid in a 25% human albumin solution. The resulting solution is freeze dried, resulting in a pale

yellow powder. The final mixture is prepared with double-distilled deionized water. Final packaging configuration and sterilization processing conditions have yet to be determined.

Table 1 details the HeatSYNC® Gel materials of construction.

Table 1. HeatSYNC® Gel Materials of Construction

Compound	Description	Amount per mL of HeatSYNC® Gel
ALBUMIN (HUMAN) 25% ⁱ	Composition of ALBUMIN (HUMAN) 25%	
	Component	Quantity/1000 mL
	Protein (≥ 96% human albumin)	250 g
	Sodium	130-160 mmol
	Potassium	≤ 2 mmol
	N-acetyl-DL-tryptophan	0.064-0.096 mmol/g protein
	Caprylic acid	0.064-0.096 mmol/g protein
	Water for Injection	ad. 1000 mL
Citric acid trisodium salt dihydrate ⁱⁱ	W302600 USP, CAS Number 6132-04-3	0.184 g
Citric acid ⁱⁱ	S1804 USP, CAS Number 77-92-9	0.0663 g
Water ⁱⁱⁱ	Double distilled, deionized	1.00 mL

ⁱ Octapharma; ALBUMIN (HUMAN) 25%; contains no preservatives.

ⁱⁱ Sigma-Aldrich

ⁱⁱⁱ Used in final step to solubilize lyophilized protein product.

Device Characterization

The extent of the required biological evaluation for a medical device depends on the nature and duration of body contact with the medical device as well as on the amount and quality of relevant material information and safety data available. HeatSYNC® Gel is injected into target tissue immediately before thermal ablation. During the ablation procedure, HeatSYNC® Gel is denatured along with the ablated tissue and remains in the body as the wound resolves. Further, non-clinical testing to date indicates that some residual gel may remain at the target site, particularly at areas of the ablation zone farthest removed from the antenna. Therefore, when evaluated in accordance with ISO 10993-1:2018 and FDA's guidance document, *Use of International Standard ISO 10993-1, "Biological evaluation of medical devices – Part 1: Evaluation and testing within a risk management process"* (September 4, 2020)^A, HeatSYNC® Gel is conservatively characterized as a tissue/bone implant with long-term contact duration (> 30 days, Type C).

ISO 10993-1:2018 gives guidance on which endpoints in the ISO 10993 series should be considered for a given application. Table 2 summarizes the endpoint recommendations of ISO 10993-1:2018 and FDA's guidance for HeatSYNC® Gel:

Table 2. Endpoints to be considered in a biological risk assessment for the Final HeatSYNC® Gel

Endpoints Recommended	Applicable ISO 10993 Standard
Physical & Chemical Information	ISO 10993-17: Establishment of allowable limits for leachable substances ISO 10993-18: Chemical characterization of materials

^A <https://www.fda.gov/media/85865/download>

Endpoints Recommended	Applicable ISO 10993 Standard
Degradation	ISO 10993-9: Framework for identification and quantification of potential degradation products
Cytotoxicity	ISO 10993-5: Tests for <i>in vitro</i> cytotoxicity
Sensitization	ISO 10993-10: Tests for irritation and skin sensitization
Irritation/Intracutaneous Reactivity ⁱ	ISO 10993-10: Tests for irritation and skin sensitization
Acute Systemic Toxicity	ISO 10993-11: Tests for systemic toxicity
Material-Mediated Pyrogenicity	ISO 10993-11: Tests for systemic toxicity
Subacute/Subchronic Toxicity	ISO 10993-11: Tests for systemic toxicity
Genotoxicity	ISO 10993-3: Tests for genotoxicity, carcinogenicity & reproductive toxicity
Implantation	ISO 10993-6: Tests for local effects after implantation
Hemocompatibility ⁱⁱ	ISO 10993-4: Selection of tests for interactions with blood
Chronic Toxicity	ISO 10993-11: Tests for systemic toxicity
Carcinogenicity	ISO 10993-3: Tests for genotoxicity, carcinogenicity & reproductive toxicity
Reproductive/Developmental Toxicity ⁱⁱ	ISO 10993-3: Tests for genotoxicity, carcinogenicity & reproductive toxicity
Immunotoxicity ⁱⁱ	ISO 10993-20: Principles and methods for immunotoxicology testing of medical devices

ⁱ Please note that ISO 10993-23:2021, Tests for irritation, was recently published; the extent to which FDA may accept this standard is not yet known.

ⁱⁱ For particular medical devices, there is a possibility that it will be appropriate to include additional or fewer endpoints than indicated; see text that follows for additional explanation.

It is important to note that while all of the endpoints listed above must be considered per standards, guidance, and additional considerations (see following paragraphs), they can be addressed in a number of different ways including by testing and/or written justification/risk assessment, if relevant information is available.

While not specifically addressed in Annex A, ISO 10993-1:2018 provides guidance for immunotoxicity testing in section 6.3.2.15, which states,

“Immunotoxicity testing shall be considered based on the chemical nature of the materials of manufacture and data from sources that are suggestive of immunotoxicological effects or if the immunogenic potential of any of the chemicals is unknown. If immunotoxicity testing is performed, it shall be conducted in accordance with ISO/TS 10993-20.”

The subject device is a protein-based product. As the likelihood of being immunogenic is increased with proteins, and the package insert^B for the Octapharma ALBUMIN (HUMAN) 25% product indicates that the most common adverse reaction observed for ALBUMIN HUMAN (any strength) during post-approval use was anaphylactic shock, the potential risk of immunotoxicity (ISO 10993-20) should be addressed. See *Risk Analysis of Device Materials* and *Justification for Testing Not Recommended*. It may be possible to address this endpoint in part with judicious selection of study measures to be collected in the planned GLP animal study.

Furthermore, as HeatSYNC® Gel could come into direct contact with the vasculature and lymphatic system when used as intended, and due to experience with FDA feedback on prior similar products, Hemocompatibility (ISO 10993-4) should also be a consideration of the risk assessment and evaluation plan. In addition, considering the long-term duration of exposure to the HeatSYNC® Gel and the nature of the materials used, the endpoints of Biodegradation (ISO

^B <https://services-prod-us.octapharma.com/download/x/224c5f275a/albumin25-pi.PDF>

10993-9), and Reproductive/Developmental Toxicity (ISO 10993-3) should also be addressed. Here too it may be possible to address these endpoints in part with selection of appropriate study measures to be collected in the planned GLP animal study.

Evaluation Approach

The biological evaluation of a medical device (or a material component of a medical device) is properly performed within the framework of a risk management process to determine the acceptability of any potential adverse biological response resulting from contact with the device. The biological safety assessment process is designed to identify potential risks associated with the clinical use of the finished device. Testing strategies and/or waiving of tests herein were developed using logical and scientifically reasoned plans for evaluating biological safety that demonstrate all biological hazards have been considered and relevant risks controlled. The biological risks deriving from use of a medical device are related to two main sources, regardless of body contact type or duration:

- Toxicological hazards arising from chemical constituents that lead to toxicological risks. These can arise from (1) the raw materials used to fabricate the device which may contain chemicals capable of leaching from the device and/or producing degradation products that cause adverse effects once in contact with the body or (2) the manufacturing process, including assembly, cleaning, packaging and sterilization which may all provide bioavailable residues.
- Materials hazards arising from incompatibility between materials and tissues that lead to an inappropriate host response.

To evaluate the biological safety of the HeatSYNC® Gel, consideration was given to the following: type of patient contact (discussed previously in the Device Characterization section), potential hazards of the materials of construction, as well as information from clinical experience with similar approved uses of the materials of construction and relevant information available in the literature.

Evaluation of Physical and Chemical Endpoints

Physical & Chemical Information

As stated in ISO 10993-1:2018, gathering physical and chemical information on a medical device (or a component of the device) is a crucial first step in the biological evaluation and its associated process of material characterization. The characterization should address the constituent chemicals of the medical device as well as possible residual process aids or additives used in its manufacture. In addition, it is possible that some physical characterization information will be needed for implanted medical devices.

It is recommended that material characterization be performed in accordance with ISO 10993-18:2020, *Biological evaluation of medical devices – Part 18: Chemical characterization of medical device materials within a risk management process*. Identification of the materials and processes used in the manufacture of a medical device enables the intrinsic toxicity of the device to be evaluated. Toxicological risk assessment methods in accordance with ISO 10993-17, *Establishment of allowable limits for leachable substances*, can address biomaterials toxicology risk associated with exposure to physiologically mobile chemical

compounds that may originate from the device. Such testing and analysis should be performed on final, finished devices once manufacturing, packaging, and sterilization conditions have been determined for the final product. If testing is performed on the device prior to establishment of these parameters, then a robust rationale for biological equivalency of the test article used compared to the final product should be developed per ISO 10993-18 requirements. Annex C of this standard outlines the concept of biological equivalence.

The chemical characterization testing plan should include the currently standard complement of analytical methods: HSGC-MS, GC-MS, LC-MS, and ICP-MS to determine extracted volatile, semi-volatile, non-volatile, and elemental substances. Exhaustive extractions on the finished, sterilized device should be conducted using polar, non-polar, and mid polar extraction solvents, unless justified otherwise. Given the nature of the product as a gel, alternative methods of sample preparation may be necessary (e.g., dissolution as compared to extraction). Feasibility testing may help to establish the most appropriate approach. See also the discussion in the section on Samples for Biological Testing. In any case, samples for analytical testing should be prepared and processed so as to provide appropriate, toxicologically relevant sensitivity. Additional factors recommended include use of a minimum of five reference standards in each method and demonstration of adequate compound recovery (e.g., via compound spiking) through any extract work-up processes.

Degradation

An assessment of potential biocompatibility risk should include physical characteristics that might contribute to an unwanted tissue response. When evaluating a material intended for implantation, potential sources for degradation should be considered. ISO 10993-9 *Biological evaluation of medical devices – Part 9: Framework for identification and quantification of potential degradation products* provides general principles for the systematic evaluation of the potential and observed biodegradation of medical devices and for the design and performance of biodegradation studies. Consideration of the possible need for degradation studies is provided in Annex A, which states:

“A.1 Degradation studies shall be considered if

- a) the device is designed to be bioresorbable or
- b) the device is intended to be implanted for longer than thirty days or
- c) an informed consideration of the material(s) system indicates that toxic substances can be released during body contact.

A.2 Degradation studies need not be necessary if

- a) the probable degradation products are the same substances, in the predicted quantities, and produced at a similar rate and in comparable location to those that are produced by devices that have a history of safe clinical use and/or
- b) the probable degradation products are particulate and are in a physical state, i.e. size, distribution and shape, and in the predicted quantities, and produced at a similar rate and in comparable location to those that are produced by devices that have a history of safe clinical use or
- c) sufficient degradation data relevant to the substances and degradation products in the intended use already exist.”

The approach to the assessment of degradation depends on the material under consideration. HeatSYNC® Gel is not a polymeric chemical product or an orthopedic implant subject to wear considerations, but a protein-based gel that will be injected into soft tissue during a thermal

ablation procedure. The mechanisms of degradation under these conditions should be described, and the risk assessed. At a high level, the protein-based gel will be denatured along with the tissue in the necrotic, ablative zone; it is expected that any remaining device material at the outside margin of the coagulated zone will resorb as the post-ablative wound heals over several weeks. Monitoring temperatures the tissues are exposed to during the feasibility animal study can inform *in vitro* conditions for forced degradation under simulated conditions and subsequent material characterization of degradative samples.

The planned non-clinical GLP animal study will further support how the ablative zone resolves over time and how long it takes to achieve a stable, non-toxic presence in the tissue (for both coagulated gel and any theoretical remaining uncoagulated gel at the periphery of the ablative zone farthest from the antennae). Multiple time points can assess the degradation *in vivo* and the biological response to the degradation of the protein-based gel over time to demonstrate that only the probable products of degradation are present in the expected quantities. Additional evaluation of degradation may be available from analytical chemistry studies; for example, studies in which degradation is forced by exposure to light or chemical agents such as acids, bases, or oxidizing agents could be considered to better characterize the potential risks of degradation.

Particulates

When a medical device has contact with the vasculature, the agency recommends measuring the size and quantity of particulates that a device may generate as an indication of the potential embolic risk. In a DRAFT guidance document, *Peripheral Percutaneous Transluminal Angioplasty (PTA) and Specialty Catheters – Premarket Notification (510(k)) Submissions (January 13, 2020)*^C regarding a Class II medical device in contact with the vasculature, the FDA recommends reporting the total quantity and size of the particulates generated during worst-case simulated use of the device in three distinct size ranges ($\geq 10 \mu\text{m}$, $\geq 25 \mu\text{m}$, and at a minimum, $\geq 50 \mu\text{m}$) to comprehensively report on any subvisible particulate matter present.

If significant particulate matter is introduced into the bloodstream during a medical procedure, it may present an off-target embolic risk to the patient. The agency expressed this concern in their written feedback to Theromic's pre-submission, Q181908:

"Your device could come into direct contact with the vasculature and/or lymphatics which introduces specific risks (i.e. hemocompatibility, thromboembolism, lymphatic blockage). Your studies should be designed to help understand and mitigate these potential risks."

In the above referenced draft guidance, the agency states:

"Particulate testing should be conducted as part of your design verification testing and should not be for characterization only. A rationale for the criteria used as well as a discussion of the results should also be provided. The discussion of acceptable particulate evaluation should include a justification that the number and size of particulates will not impact safety or clinical performance.

...If large amounts of particulates are shed, it may be important to demonstrate comparability to a legally-marketed predicate device used in the same target vasculature or provide evidence of safety through your animal studies (with appropriate downstream assessments). A scientific rationale should be provided to support the particulate acceptance criteria that is used."

^C <https://www.fda.gov/media/134016/download>

To mitigate the potential risk of particulate matter embolism and to demonstrate adequate control of manufacturing processes, an analysis of particulates in the final, finished product is recommended to further characterize the HeatSYNC® Gel with the above guidance in mind. An option would be testing in accordance with one of the methods described in USP <788>, “*Particulate Matter in Injections.*”^D The particulate analysis procedures described provide criteria for solutions for injections and parenteral administration; while the USP chapter does not provide acceptable thresholds for particles generated by medical devices, the acceptance criteria for injected solutions may be accepted with adequate justification. The planned non-clinical GLP animal study can further support adequate mitigation of embolic risk.

Risk Analysis of Device Materials

A literature review on the materials used to construct a permanent implant is part of a comprehensive risk analysis approach. In order to identify relevant toxicity data on the materials used to construct the HeatSYNC® Gel, multiple sources were searched. These sources included on-line databases, such as ChemIDplus. ChemIDPlus indexes dozens of relevant toxicity databases such as ATSDR, CCRIS, DART, EMIC, GENETOX, HSDB, IARC, IRIS, MEDLINE, NTPA, RTECS, TSCAINV (the Toxic Substances Control Act Inventory), and TOXLINE. Additional sources examined included ChemFinder and the World Wide Web using appropriate search engines. The search terms included elements such as CAS numbers (when available), chemical names, chemistry, toxicology, toxicity or biocompatibility. The literature review on each material is not intended to be exhaustive as presented in the guidelines of the informative Annex C of ISO 10993-1 but to provide information on actual hazards related to raw materials. To assess the overall toxicological risks, other parameters must also be taken into account such as the manufacturing process (including packaging and sterilization) and its potential residues. The intent of this section is to preliminarily identify any documented toxicological hazards as potential risks.

- Human Serum Albumin

Human serum albumin is the most abundant protein in human plasma and helps maintain osmotic pressure and pH in the blood stream. It is a non-glycosylated monomeric protein responsible for several biological functions. Albumin maintains colloid osmotic pressure and serves as a transport protein for various fatty acids, growth factors, chaperones, metal ions, and toxic substances. The normal concentration of albumin in blood is approximately 35-50 g/L. Albumin is used to replace blood volume loss resulting from trauma such as severe burns or an injury that causes blood loss. Albumin can also be administered to treat low albumin levels caused by various diseases. Hypoalbuminemia is often observed in patients with liver disease, renal failure, nephrotic syndrome, hyperthyroidism, and malnutrition.¹ High doses of albumin are typically required for clinical applications, typically in excess of 10 g/dose.² In addition to its use as a colloid replacement solution, albumin is used as an excipient and to increase the stability of other therapeutics.¹

^D Pharmacopeia, U. S., 2011, “Particulate Matter in Injections,” The United States Pharmacopeial Convention, United States Pharmacopeia, Rockville, MD, p. 788.

Human albumin solution is available for therapeutic medical use, usually at concentrations of 5-25%. Human albumin can be produced by recombinant methods; however, it is most commonly purified from human plasma. The main source of human serum albumin for clinical applications is from donated human blood and, therefore, imposes a risk to end users of being exposed to pathogenic materials.³ While there is no strong medical evidence that albumin administration (compared to saline) offers a mortality benefit for people who have hypovolemia or for those who are critically ill due to burns or hypoalbuminemia,⁴ human-derived albumin has been used clinically in thousands of patients with an excellent safety profile. Adverse reactions to human serum albumin are rare, but potentially life-threatening.⁵

The human serum albumin used in the HeatSYNC® Gel is a sterile, liquid preparation of albumin derived from large pools of human plasma. While no procedure has been shown to be completely effective in removing viral infectivity from derivations of human plasma, the manufacturing processes used (i.e., pasteurization at 60 °C for several hours) have provided a significant viral reduction in *in vitro* studies.³ During pasteurization and storage of albumin blood products, N-acetyl-DL-tryptophan and caprylic acid are used as an antioxidant and thermal stabilizer for albumin, respectively. These stabilizers are discussed further below.

- N-acetyl-DL-tryptophan

Biotherapeutics can be susceptible to oxidation during manufacturing and storage.

N-acetyl-DL-tryptophan has been used for decades as a stabilizer in protein solutions and is included in parenterally-administered drugs approved in the US, EU, and Japan.⁶

N-acetyl-DL-tryptophan appears in the FDA's Inactive Ingredient List for parenteral formulations and has a history of safe use without identification of hazard in acute settings.⁷

- Caprylic Acid

Caprylic acid is the chemical name for octanoic acid, a naturally occurring fatty acid. It is found in the milk of certain mammals as well as coconut oil and palm oil; it is a medium-chain fatty acid. Caprylic acid has antimicrobial and anti-inflammatory properties and is found in dietary supplement form; it is a generally recognized as safe (GRAS) direct food additive, restricted only by the need to follow good manufacturing practices (21 CFR 184.1025).

Caprylic acid, taken either by mouth or used topically, is thought to treat a number of health conditions, including: Alzheimer's disease, epilepsy, skin infections, hyperlipidemia, hypertension, irritable bowel disease, hypoalbuminemia, and lactose intolerance. Caprylic acid has been shown to eradicate Gram-positive, Gram-negative, and fungal biofilms as a biocompatible, non-antibiotic antimicrobial.^{8,9} It is used intravenously in some total parenteral nutrition formulations and has demonstrated anticonvulsant effects in mice.¹⁰ Caprylic acid is also widely used in the manufacturing of sanitizers, disinfectants, perfumes, anti-aging creams, and dyes.

- Citric Acid (CAS 77-92-9)/ Sodium Citrate (CAS 6132-04-3)

Citric acid (2-hydroxy-1,2,3-propanetricarboxylic acid) and its salts are solids. Citric acid is soluble in water and in some organic liquids and is very hydrophilic. Citric acid is triprotic and thus has 3 distinct pK_as, making it an excellent buffer component. Even for the most

acidic of the carboxylate functional groups, citric acid is only a weak acid with a pK_a of 3.1. The chemical structures and available data indicate that the physical-chemical properties, environmental fate behavior, and aquatic and mammalian toxicity of citric acid and sodium citrate are similar.¹¹ Industrial, large-scale production of citric acid is most commonly accomplished by fermentation of crude sugar stocks. The citrate salts are produced by the same fermentation processes, but are crystallized in the presence of appropriate alkaline solutions (e.g., citric acid can be crystallized with sodium hydroxide to produce sodium citrate).

Citric acid has a vital function in human and animal metabolism. It appears as an intermediate in the basic physiological citric acid cycle (Krebs cycle) in every eukaryotic cell. In this metabolic pathway, citric acid completes the breakdown of pyruvate, formed from glucose through glycolysis, and it liberates carbon dioxide. Approximately 2 kg of citric acid are formed and metabolized every day in the average human.¹² Citric acid is widely used in the foods/beverages, pharmaceutical/chemical, textile, and electroplating industries.¹³ It is employed as a preservative, emulsifier, acidulant, flavoring, and buffering agent in pharmaceuticals as well as foods; citric acid also serves as an excellent, harmless disinfectant against several viruses.

Orally administered citric acid is well absorbed and largely metabolized. As listed by the FDA, citric acid and sodium citrate are both generally recognized as safe (GRAS) direct food additives, restricted only by the need to follow good manufacturing practices, thereby establishing these compounds as safe for ingestion (21 CFR 184.1033). The safety of citric acid in topical applications has been reviewed by the Cosmetic Ingredient Review (CIR) Expert Panel.¹² In its guidance, *Labeling for Cosmetics Containing Alpha Hydroxy Acids* (January, 2005)^E, the FDA specifically mentions citric acid-containing products. Additionally, for the last two decades, citric acid has been applied to cross-linking of different other materials including ultrafine protein fibers for biomedical applications.

Based on many experimental data in animals and on human experience, citric acid is of low acute toxicity.¹¹ The NOAEL for repeated dose toxicity for rats is 1,200 mg/kg/d. The major, reversible (sub)chronic toxic effects seem to be limited to changes in blood chemistry and metal absorption/excretion kinetics. Citric acid is not suspected of being a carcinogen nor a teratogenic agent. The NOAEL for reproductive toxicity for rats is 2,500 mg/kg/d. Further, it is not mutagenic *in vitro* nor *in vivo* and its sensitizing potential is seen as low. In contrast, irritation, in particular of the eyes but also of the respiratory pathways and the skin, is the major toxicological hazard presented by citric acid; this conclusion is confirmed by a series of reports relating to eye and skin irritation.

An analysis of the materials used to construct HeatSYNC® Gel indicates that all are well characterized and do not present any toxicological hazard. The materials themselves are naturally occurring and well tolerated at high doses. It can be concluded that the materials of construction of the HeatSYNC® Gel are suitable in terms of biological safety. Potential chemical degradation

^E [guidance-industry-labeling-cosmetics-containing-alpha-hydroxy-acids](#)

post-ablation and any associated risks will be identified through simulated ablation at the bench, material characterization of degradative samples, and non-clinical animal testing.

Evaluation of Biological Endpoints

Samples for Biological Testing

FDA recommends testing medical devices in the condition they will be used. As stated previously for evaluation of physical and chemical endpoints, testing for the evaluation of biological endpoints should be performed on final, finished devices once manufacturing, packaging, and sterilization conditions have been determined for the final product. If testing is performed on the device prior to establishment of these parameters, then a robust rationale for biological equivalency of the test article used compared to the final product should be developed per ISO 10993-18 requirements. Annex C of this standard outlines the concept of biological equivalence.

While extractions for chemical characterization are covered in ISO 10993-18, guidance on the procedures in the preparation of samples and the selection of reference materials for medical device testing in *biological test systems* is provided in ISO 10993-12:2021, *Biological Evaluation of Medical Devices - Part 12: Sample Preparation and Reference Material*. Test sample selection is covered in clause 7, which states:

“Testing shall be performed on the final product, representative samples from the final product, materials processed in the same manner as the final product (see ISO 10993-1), or on appropriate extracts of any of these. The choice of test sample shall be justified.

NOTE In the case of materials that cure *in situ*, different test samples representative of the cured material versus the uncured state of the material might be needed.

For absorbable materials that could potentially have toxic degradants and residuals, testing of intermediate products should be considered.

The same test sample selection procedure applies when an extract is required.”

Generally, the use of both aqueous and organic extraction solvents is a requirement of Part 12. A pilot study should be considered to determine the compatibility of the HeatSYNC® Gel with the extraction solvents and temperatures required for ISO 10993 testing, unless already assessed. Assessment of compatibility with the required solvent polarities should consider signs of particulates and overall clarity of the extraction. See also the discussion of sample preparation in the section Physical & Chemical Information.

Appropriate extractions of materials representative of the final product should be carried out using an extraction ratio of 0.2 g/mL for an irregularly shaped solid device in alignment with ISO 10993-12. Alternatively, if the specific ISO 10993 test allows, the HeatSYNC® Gel may be applied to the test system directly. The relevant text is found in clause 10.3.11(a-c) of Part 12, which states that for solution and soluble materials, the standard extraction methods used for insoluble materials might be inappropriate under the following conditions:

“a) Factors such as test system compatibility, route of administration and extent of dissolution or degradation should be considered in the final preparation for testing. Use an appropriate vehicle and conditions to simulate exaggerated exposure wherever possible. A pre-test can help to determine appropriate conditions.

b) If the material completely dissolves, in a vehicle or diluent that is compatible with the material and the test system, the resulting solution can be evaluated neat, provided the solution properties are also compatible with the test system, e.g. pH, osmolarity, solute concentrations. If the resulting solution contains all the constituents of the material, a second vehicle would not be needed.

c) If the material is an aqueous solution and used in this form, it shall be tested directly and not extracted, provided the solution properties are compatible with the test system [see also a) and b) above].”

In light of this, and dependent upon the individual test system, two different extraction vehicles may not be needed for all biological safety tests that generally make use of both polar and non-polar extracts if HeatSYNC® Gel can be applied to the test system directly. Additionally, if viscosity is the only challenge to the test system, an aqueous dilution of the gel may suffice as a single extract with justification as the resulting solution would contain all the constituents of the material.

Biocompatibility Testing Recommended

A summary of the methods recommended to address each relevant biological endpoint is provided in Table 3. Justifications for testing not recommended are presented after the table.

Table 3. Recommended method to address biological endpoints for the final HeatSYNC® Gel

Endpoints Recommended	Method to Address the Endpoint
Cytotoxicity	L929 MEM Elution per ISO 10993-5
Sensitization	Guinea Pig Maximization (2 extracts ⁱⁱ) per ISO 10993-10
Irritation/Intracutaneous Reactivity ⁱ	Intracutaneous Reactivity (2 extracts ⁱⁱ) per ISO 10993-10
Acute Systemic Toxicity	Acute Systemic Toxicity in mice (2 extracts ⁱⁱ) per ISO 10993-11
Material-Mediated Pyrogenicity	Material Mediated Pyrogenicity per ISO 10993-11 / USP <151>
Subacute/Subchronic Toxicity	<i>Address with rationale</i>
Genotoxicity	AMES Bacterial Reverse Mutation (2 extracts ⁱⁱ) per ISO 10993-3
	Mouse Lymphoma Assay (2 extracts ⁱⁱ) per ISO 10993-3
Implantation	Implantation in rabbit (4 wk, 13 wk) per ISO 10993-6 / alternatively, may consider evaluating in GLP animal study
Hemocompatibility	Material-mediated hemolysis per ASTM F756 and ISO 10993-4
	Complement activation (Sc5b9) per ISO 10993-4
Chronic Toxicity	<i>Address with rationale</i>
Carcinogenicity	<i>Address with rationale</i>
Reproductive/Developmental Toxicity	<i>Address with rationale</i>
Immunotoxicity	<i>Address with rationale</i>

ⁱ Please note that ISO 10993-23:2021, Tests for irritation, was recently published; the extent to which FDA may accept this standard is not yet known.

ⁱⁱ Alternatively, testing the HeatSYNC® Gel neat or as a single polar extract may suffice with justification if compatible with the test system as the test article would contain all constituents of the material.

Please note the following considerations regarding the ISO 10993 tests recommended in Table 3:

- **Intracutaneous Reactivity**

Traditionally, small animal tests are performed prior to testing on humans to help predict human response. More recently, *in vitro* tests have been added as adjuncts or alternatives. *In vitro* tests have preference over *in vivo* tests when appropriately validated and providing equally relevant information to that obtained from *in vivo* tests. The recently published ISO 10993-23 requires use of an *in vitro* model as a step-wise approach prior to resorting to animal use and establishes the equivalency of the *in vitro* test for assessing this endpoint; the relevant text is found in clause 6.1:

“The *in vitro* method with RhE models for testing irritation was developed specifically to detect skin irritation potential for neat chemicals (see OECD 439). The method was adapted and validated with two RhE models for detection of irritant chemicals in medical device extracts. This method was found equally sensitive to detect low concentrations of some strong irritant in extracts from polymeric medical materials (PVC and silicone) when compared to the human patch testing and intracutaneous rabbit test. Hence, the

RhE test as described in this document can replace the *in vivo* rabbit test for irritation by skin exposure and by intracutaneous (intradermal) administration.”

However, the extent to which FDA may recognize this new standard is not yet known. The introduction of ISO 10993-10 *Biological evaluation of medical devices – Part 10: Tests for irritation and skin sensitization*, states:

“Despite progress and considerable effort in this direction, a review of findings suggest that currently no satisfactory *in vitro* test has been devised to eliminate the requirement of *in vivo* testing. Where appropriate, the preliminary use of *in vitro* methods is encouraged for screening purposes prior to animal testing. In order to reduce the number of animals used, this part of ISO 10993 presents a step-wise approach, with review and analysis of test results at each stage. An animal test is usually required prior to human testing,”

Clause 4 of this agency-recognized standard notes:

“For medical devices that are used as implants or external communicating devices, intradermal testing is more relevant in approaching the application and so for detection of irritation activity, intracutaneous testing shall be used...”

For medical devices that are contacting breached or compromised surface, externally communicating or used as an implant, the use of an intracutaneous (intradermal) reactivity test is indicated. Therefore, it is recommended that the Intracutaneous Reactivity test per ISO 10993-10 be applied in this case.

- **Acute Systemic Toxicity**

The 10993-11 Acute Systemic Toxicity in mice is recommended. Typically, this test is performed using polar and nonpolar extracts of the device. If compatible with the test system, HeatSYNC® Gel may be used directly. In this case, the amount injected should be in accordance with Part 11; guidance on dosage volume is provided in Annex B. Additional support for this endpoint could potentially be leveraged from the planned large animal study.

- **Genotoxicity**

No single test is capable of detecting all relevant genotoxic agents; therefore, a battery of tests is usually recommended. FDA’s guidance document, Use of International Standard ISO 10993-1, “*Biological evaluation of medical devices – Part 1: Evaluation and testing within a risk management process*” (September 4, 2020) discusses genotoxicity test selection and recommends the Bacterial Reverse Mutation Assay (AMES Test) and an *in vitro* mammalian genotoxicity assay be performed. An optional *in vivo* cytogenetics assay, such as a bone marrow chromosomal aberration assay, is also recommended for devices containing novel materials. The relevant text is found in Section VI., Part F, *Genotoxicity*:

“Because no single test can detect all genotoxins, we recommend the following two *in vitro* tests be conducted, as well as an optional third *in vivo* test:

- Bacterial gene mutation assay. This test is conducted with engineered strains of Salmonella typhimurium and Escherichia coli designed to detect all possible single base pair changes as well as frameshift mutations [OECD 471 (1997) “Guidelines for Testing of Chemicals – Bacterial Reverse Mutation Test”].
- An *in vitro* mammalian genotoxicity assay. A choice of one of the following is recommended:
 - a) the mouse lymphoma gene mutation assay [OECD 476 (1997) “Guidelines for the Testing of Chemicals – In Vitro Mammalian Cell Gene Mutation Test”], which is preferred since it detects the broadest set of genotoxic mechanisms associated with carcinogenic activity;

- b) an *in vitro* chromosomal aberration (CA) assay [OECD 473 (2014) “Guidelines for the Testing of Chemicals – In Vitro Mammalian Chromosome Aberration Test”]; or
- c) an *in vitro* micronucleus assay [OECD 487 (2014) “Guidelines for the Testing of Chemicals – In Vitro Mammalian Cell Micronucleus Test”].
- An *in vivo* cytogenetics assay should be considered, for example, for devices containing novel materials. However, if the quantities of materials in the test extract following exhaustive extraction of the devices are below the threshold of detection of the *in vivo* assay, the test does not need to be performed. When an *in vivo* assay is needed, a choice of one of the following is recommended:
 - a) a bone marrow micronucleus (MN) Assay [OECD 474 (2014) “Guidelines for the Testing of Chemicals – Mammalian Erythrocyte Micronucleus Test”]; or
 - b) a bone marrow chromosomal aberration (CA) assay [OECD 475 (1997) “Guidelines for the Testing of Chemicals – Mammalian Bone Marrow Chromosome Aberration Test”]; or
 - c) a peripheral blood MN assay (OECD 474).”

The Mouse Lymphoma Assay is recommended in Table 3 as it is capable of detecting the broadest set of genotoxic mechanisms associated with carcinogenic activity, but any of the *in vitro* mammalian genotoxicity assays would suit; the Chromosomal Aberration Assay is often employed. Clause 5.2.2 of Part 3 notes that “An *in vivo* assay is not necessary, if the user can demonstrate that the quantities of extractables from the test article are less than the amount of material that would induce a positive response with a potent well-characterized *in vivo* micronucleus genotoxin.” Although the HeatSYNC® Gel is a novel product, it is a protein-based gel that contains no known toxicological hazard. Therefore, an *in vivo* genotoxicity test is not warranted unless indicated by future chemical characterization of the gel and/or its degradative products.

- **Implantation**

For long term implant devices such as the HeatSYNC® Gel, the implantation endpoint should be addressed. The recommended ISO 10993-6 implantation in rabbit should include Part 11 endpoints for toxicity and should also evaluate degradation of the material over time. Interim assessments of the implanted material should assess the tissue response during degradation. This study can inform the planned GLP pivotal animal study endpoints and further characterize the material. Depending on timing and schedules, it could also be possible to design the large animal study in such a way as to address the implantation endpoint without the need for the traditional rabbit study. Such an approach would entail collecting the Part 6 endpoints within the large animal study.

- **Hemocompatibility**

Recommended tests for devices that may have contact with circulating blood are found in ISO 10993-4 *Biological evaluation of medical devices – Part 4: Selection of tests for interactions with blood*, Table 1, and are indicated based on device type and patient contact. Given that the HeatSYNC® Gel is an implant and may come into contact with the vasculature during use, the potential to cause material-induced hemolysis and thrombogenicity should be evaluated in testing if not otherwise addressed during the risk assessment process. Additionally, a footnote to Table 1 indicates that complement activation should be considered for other endpoints such as anaphylaxis. Hemolysis of red blood cells is assessed using *in vitro* tests. Direct methods determine hemolysis due to physical and chemical interactions with red blood cells. Indirect methods determine hemolysis due to extractions

from test articles. The red blood cell membrane is a semi-permeable membrane susceptible to osmotic pressure-induced hemolysis and test selection should be based upon the device and the conditions in which it will be used. It is recommended that hemocompatibility be addressed with either a direct or indirect method for material/surface-mediated hemolysis per ASTM F756, depending on appropriateness of the interaction of the gel with the test system as well as complement activation (Sc5b9). Given the only incidental contact with blood in use, perhaps direct only is enough. Hemocompatibility should be further assessed in the GLP pivotal large animal study and should include gross and histopathological evaluation of local as well as downstream tissues for evidence of thrombus.

Justification for Testing Not Recommended

Clause 4.2 of ISO 10993-2 *Biological evaluation of medical devices – Part 2: Animal welfare requirements*, applies to any *in vivo* testing being considered. For the purpose of the 10993 series, animal tests shall only be justified when:

- “– the resulting data are not otherwise available, but are essential to properly characterize the test material in the context in which it is to be used;
- when no suitable scientifically validated test method not involving the use of living animals is reasonably and practically available;
- when relevant reduction and refinement strategies have been identified and implemented including, if appropriate, obtaining test data from manufacturers and suppliers, and literature searches for toxicity and biocompatibility data.”

The standard further clarifies that prior to evaluating the biocompatibility of device materials in animals, a review of available and relevant information on the properties of the test material shall be undertaken and documented; animal tests are deemed necessary only when:

- “a) they have been shown to be relevant and reliable for the purposes for which they are undertaken;
- b) the resulting data are essential to properly characterize and evaluate the test material in the context in which it is to be used in medical devices;
- c) no scientifically valid test method not requiring the use of living animals is reasonably and practically available;
- d) other relevant and appropriate strategies to minimize the pain, suffering, distress and lasting harm caused to the animals that are used have been identified and implemented.”

A description of medical device chemical constituents and consideration of material characterization including chemical characterization per ISO 10993-18 should precede any biological testing. Chemical characterization with an appropriate toxicological threshold can be used to determine if further testing is needed. Justification for the omission of specific testing for Subacute/Subchronic Toxicity, Chronic Toxicity, Carcinogenicity, Reproductive/Developmental Toxicity, and Immunotoxicity of the HeatSYNC® Gel is provided below:

- **Subacute/Subchronic Toxicity / Chronic Toxicity**

Cytotoxicity testing, due to the unprotected nature of the mouse fibroblast cells (L-929) *in vitro*, is often used to screen materials intended for use in the body. Results from this screening test typically provide good correlation to the potential tissue response to a material. If the HeatSYNC® Gel successfully passes the Cytotoxicity test, it can be concluded that the materials can be regarded as potentially biologically safe. The planned chemical characterization testing and associated toxicological evaluation can further support the lack

of potential for significant patient exposure to compounds that could pose a potential risk of systemic toxicity. Together, if favorable, these results can indicate that the amounts and types of leachable chemicals from the HeatSYNC® Gel are not likely to result in adverse biological effects in clinical use of the device, and thereby obviate the need for *in vivo* chronic toxicity studies. These endpoints can be further supplemented with systemic toxicity endpoints from the planned GLP pivotal animal study with collection of the Part 6 endpoints.

- **Carcinogenicity / Reproductive/Developmental Toxicity**

Per ISO 10993-3, carcinogenicity is an endpoint that is not commonly tested for; however, it may be more necessary for novel materials. FDA guidance indicates that reproductive and developmental toxicity should be addressed for novel materials. These endpoints can be addressed with toxicological risk assessment of extractable and leachable compounds of potential exposure. The results from genotoxicity testing can further supplement the biological safety evidence for these endpoints.

- **Immunotoxicity**

HeatSYNC® Gel is a protein-based product. As the likelihood of being immunogenic is increased with proteins, and the package insert for the Octapharma ALBUMIN (HUMAN) 25% product indicates that the most common adverse reaction observed for ALBUMIN HUMAN (any strength) during post-approval use was anaphylactic shock, the immunotoxicity endpoint should be addressed. The methods for assessment of immunotoxicity and their predictive value are described in ISO 10993-20 *Biological evaluation of medical devices – Part 20: Principles and methods for immunotoxicology testing of medical devices*. The planned chemical characterization testing and associated toxicological evaluation can support the lack of potential risk of immunotoxicity. The complement system is a defensive biochemical cascade intended to clear pathogens from the body. It is a humoral response independent of antibody-specific mechanisms. The recommended complement (sC5b9) testing can assess potential for humoral immune response (anaphylaxis). Cellular immune responses can be assessed via the recommended guinea pig sensitization test. The recommended sub-acute systemic toxicity test in mice will be useful for obtaining general indications of potential immunosuppression or immunostimulation. Per ISO 10993-20, only materials that show evidence of causing immunosuppression or immunostimulation should be subjected to further investigation. Furthermore, observations during the planned GLP pivotal animal study (and implantation study in rabbits, if performed) can also be indicative of an immune response and can include clinical symptoms such as allergy, skin rash, urticarial, edema, lymphadenopathy and inflammation. Chronic inflammation associated with immunotoxicity would be histopathologically distinct (predominant in lymphocytic cells as opposed to a foreign body reaction characterized by presence of macrophages and foreign body giant cells at the tissue/material interface). Taken together, if confirmatory of biological safety, these assessments are sufficient and no further testing for this endpoint is warranted.

Summary

Considering the potential biological impact, this biological evaluation plan should be considered a living document and updated as development work and available data accrues. The interpretation of the overall biocompatibility evaluation plan should be considered in the appropriate benefit-risk context. This BEP was developed to address knowledge gaps either by

recommended testing or other evaluations that appropriately address the risks. Future product development and testing may indicate that the plan should be adjusted. Such testing and analysis should be performed on final, finished devices once final packaging configuration and sterilization processing conditions have been determined. If testing is performed on the device prior to establishment of these parameters, then a robust rationale for biological equivalency of the test article used compared to the final product per ISO 10993-18 requirements will be required.

DRAFT

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Appendix 1

Qualifications of Authors